

Passive Risk Detection Report

devdaytest1 | Harness CD | Cluster: validation-cluster | Namespace: default

Scan Identity: test-scan-1778747446 | Scanned: 2026-05-14 10:18 UTC

HARNESS RESILIENCE TESTING - EXPERIMENTAL | Confidential



This Passive Risk Detection report covers the Kubernetes workloads deployed in cluster 'validation-cluster', namespace 'default', for project 'devdaytest1'. A total of 7 risks were identified across 1 service(s), comprising 5 High, 2 Medium, and 0 Low severity findings. All 7 risks are currently Open and require immediate attention by the platform engineering team.

Scan Metrics Overview

Service Count	1
Total Risks	7
Services Affected	nginx
Chaos Coverage	0% (no tests yet)
Open Risks	7
Scan Type	kubernetes-security

Services Found

Service	Total Risks	High	Medium	Low	RRS
nginx	7	5	2	0	58

RRS = Risk Rating Score: *High* x 10 + *Medium* x 4 + *Low* x 1

Risk Detail Cards

HIGH

k8s-missing-resource-limits

The nginx container has no CPU or memory limits. It can consume all available node resources, starving sibling pods.

Affects: Deployment/nginx in namespace hce

Impact: *Unbounded resource consumption causes noisy-neighbour failures on the node. Memory hog can trigger kernel OOM kills on sibling pods.*

First Detected: 2026-05-14 10:21 UTC Last Detected: 2026-05-14 10:21 UTC

kubernetes

reliability

nginx

HIGH

k8s-single-replica

The nginx Deployment runs with replicas: 1, making it a single point of failure. A pod-delete fault causes 100% service unavailability until the pod restarts.

Affects: Deployment/nginx in namespace hce

Impact: *Complete service downtime during any pod disruption, node maintenance, or OOM kill. No redundancy to absorb pod-level failures.*

First Detected: 2026-05-14 10:18 UTC Last Detected: 2026-05-14 10:18 UTC

kubernetes

availability

nginx

MEDIUM

k8s-missing-resource-requests

The nginx container has no CPU or memory requests. It is assigned BestEffort QoS and will be the first pod evicted under any node pressure.

Affects: Deployment/nginx in namespace hce

Impact: *BestEffort QoS means nginx is evicted before any other pod on the node when memory or CPU pressure occurs -- even if nginx itself is not causing pressure.*

First Detected: 2026-05-14 10:22 UTC Last Detected: 2026-05-14 10:22 UTC

kubernetes

reliability

nginx

HIGH

k8s-missing-probes

The nginx container has no livenessProbe or readinessProbe. Kubernetes routes traffic to unhealthy pods and never restarts stuck ones.

Affects: Deployment/nginx in namespace hce

Impact: *Traffic is sent to pods that are restarting or deadlocked. Without livenessProbe, a hung nginx process is never restarted automatically.*

First Detected: 2026-05-14 10:22 UTC Last Detected: 2026-05-14 10:22 UTC

kubernetes

availability

nginx

HIGH

k8s-missing-network-policy

The nginx Deployment has no NetworkPolicy. All ingress and egress traffic is unrestricted, enabling lateral movement from any compromised pod in the cluster to reach nginx directly.

Affects: Deployment/nginx in namespace hce

Impact: *Any pod in any namespace can send traffic to nginx without restriction. A compromised workload can exfiltrate data, probe nginx endpoints, or launch attacks.*

First Detected: 2026-05-14 15:14 UTC Last Detected: 2026-05-14 15:14 UTC

kubernetes

security

network

nginx

MEDIUM

k8s-missing-pod-anti-affinity

The nginx Deployment has no podAntiAffinity or topologySpreadConstraints. All replicas may be scheduled on the same node, turning the multi-replica deployment into a node-level single point of failure.

Affects: Deployment/nginx in namespace hce

Impact: *Despite having replicas configured, all pods could land on the same node. A single node-drain or node-restart takes down every replica simultaneously.*

First Detected: 2026-05-14 10:33 UTC Last Detected: 2026-05-14 10:33 UTC

kubernetes

availability

nginx

HIGH

k8s-latest-image-tag

The nginx container uses image 'nginx' with no tag. A pod restart may pull a different version from the registry, causing silent version drift between replicas.

Affects: Deployment/nginx in namespace hce

Impact: *Non-deterministic deployments -- two pods in the same Deployment can silently run different image versions after independent restarts.*

First Detected: 2026-05-14 10:19 UTC Last Detected: 2026-05-14 10:19 UTC

kubernetes

reliability

nginx

Severity Distribution

Severity Level	Count	% of Total	Definition
High	5	71.4%	Critical issue posing immediate risk to availability, reliability, or security.
Medium	2	28.6%	Moderate issue that degrades resilience but does not cause immediate outage.
Low	0	0.0%	Minor issue with limited blast radius; should be addressed in normal sprint cycle.

Category Breakdown

Category	Count	Risks Included
Availability	0	
Reliability	0	
Security	0	

Recommendations & Next Steps

Priority	Service	Action	Experiment Type
P1	nginx	Set CPU/memory limits on containers	CPU Stress / Memory Stress
P1	nginx	Increase replicas + add PDB	Pod Delete
P2	nginx	Set CPU/memory requests on containers	Node CPU Hog / Node Memory Hog
P1	nginx	Add livenessProbe + readinessProbe	Pod Delete / Network Chaos
P1	nginx	Apply NetworkPolicy (ingress+egress)	Network Partition
P2	nginx	Add podAntiAffinity / topology spread	Node Drain
P1	nginx	Pin image to specific digest/tag	Container Kill

RECOMMENDED IMMEDIATE ACTION

Execute a Pod Delete chaos experiment on Deployment/nginx in namespace default. With replicas:1 and no PDB, this will result in 100% service unavailability until the pod restarts. Remediate by scaling to >=2 replicas and applying a PodDisruptionBudget (minAvailable:1).

AUTO-PARAMETERISED EXPERIMENT CONFIG

namespace: default
blast_radius: 1 pod (all replicas of nginx)
hypothesis: Service returns HTTP 200 within 30s of pod deletion
duration: 60s
rollback_trigger: Error rate > 5% OR latency p99 > 2s

Test Coverage Gap Analysis

Service	Chaos Tests	Load Tests	Coverage %	Notes
nginx	0	0	0%	Critical
TOTAL	0	0	0%	No coverage - CRITICAL

Appendix & Glossary

How This Report Was Generated

This Passive Risk Detection (PRD) report was generated automatically by the Harness Resilience Testing AI Agent. The agent scanned the Kubernetes cluster via the Harness Chaos Engineering platform, analysed each workload's configuration against a set of resilience best-practice rules, and classified each deviation as a risk with severity High, Medium, or Low. No live traffic was disrupted during the scan; only cluster API metadata was inspected.

Risk scores are computed as: $RRS = (High \times 10) + (Medium \times 4) + (Low \times 1)$, producing a deterministic numeric indicator for prioritisation. The report is generated in PDF format using the fpdf2 library and committed to the project's GitHub repository under results/ for audit trail purposes. All timestamps are in UTC. All data originates directly from the scan result JSON - no values are inferred or fabricated.

Glossary

Term	Definition
RRS	Risk Rating Score: High x 10 + Medium x 4 + Low x 1
PDB	PodDisruptionBudget: Kubernetes policy limiting simultaneous pod evictions
HPA	HorizontalPodAutoscaler: Kubernetes controller for automatic replica scaling
MTTR	Mean Time To Recovery: Average time to restore a service after an incident
PRD-Report	Passive Risk Detection Report: This document
HIGH	Severity requiring immediate remediation; poses direct risk to uptime or security
MEDIUM	Severity degrading resilience; should be resolved within the current sprint
LOW	Severity with limited blast radius; address in normal backlog prioritisation